1.- Ordena de mayor a menor estas fracciones:

$$\frac{7}{12},\ \frac{4}{6},\ \frac{5}{9},\ \frac{3}{4},\ \frac{13}{18}$$

$$m.c.m.(12,6,9,4,18) = 3^2 \cdot 2^2 = 36$$

$$\frac{7}{12} = \frac{21}{36} \qquad \frac{4}{6} = \frac{24}{36} \qquad \frac{5}{9} = \frac{20}{36} \qquad \frac{3}{4} = \frac{27}{36} \qquad \frac{13}{18} = \frac{26}{36}$$

$$\frac{27}{18} > \frac{26}{36} > \frac{24}{36} > \frac{21}{36} > \frac{20}{36}$$

$$\frac{3}{4} > \frac{13}{18} > \frac{4}{6} > \frac{7}{12} > \frac{5}{9}$$

2.- Agrupa, entre las siguientes fracciones, las que sean equivalentes:

$$\frac{10}{15} \quad \frac{5}{7} \quad \frac{1}{3} \quad \frac{5}{15} \quad \frac{2}{3} \quad \frac{2}{6} \quad \frac{15}{21}$$

$$\frac{10}{15} = \frac{2}{3};\quad \frac{5}{7} = \frac{15}{21};\quad \frac{1}{3} = \frac{5}{15} = \frac{2}{6}$$

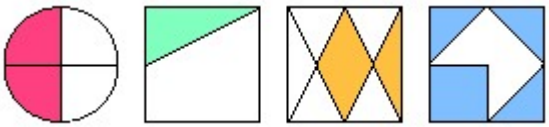
3.- Simplifica:

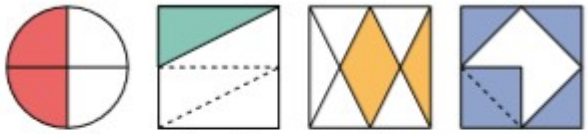
a) $\frac{30}{42}$ b) $\frac{18}{72}$ c) $\frac{75}{125}$ d) $\frac{60}{210}$ e) $\frac{2000}{4000}$

a) $\frac{30}{42} = \frac{5}{7}$ b) $\frac{18}{72} = \frac{1}{4}$ c) $\frac{75}{125} = \frac{3}{5}$

d) $\frac{60}{210} = \frac{2}{7}$ e) $\frac{2\,000}{4\,000} = \frac{1}{2}$

4.- Escribe la fracción que representa la parte coloreada en cada una de estas figuras y ordénalas.





$\frac{1}{2} = \frac{4}{8}$ $\frac{1}{4} = \frac{2}{8}$ $\frac{3}{8}$ $\frac{5}{8}$

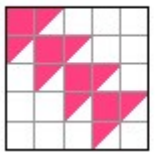
$\frac{1}{4} < \frac{3}{8} < \frac{1}{2} < \frac{5}{8}$

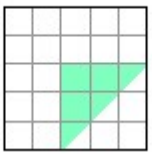
5.- Escribe una fracción equivalente a 2/5 y otra equivalente a 7/6, pero que tengan el mismo denominador.

m.c.m. (5, 6) = 30 $\frac{2}{5} = \frac{12}{30}$; $\frac{7}{6} = \frac{35}{30}$

6.- Expresa en forma de fracción la parte coloreada de estas figuras:

a) b) c)







a) $\frac{8}{25} = 0{,}32$ b) $\frac{9}{50} = 0{,}18$ c) $\frac{17}{25} = 0{,}68$

7.- Ordena las fracciones de menor a mayor:

$\frac{13}{20}$, $\frac{14}{25}$ y $\frac{7}{10}$.

1ª forma: Expresamos las fracciones en forma decimal:

$\frac{13}{20} = 0{,}65$ $\frac{14}{25} = 0{,}56$ $\frac{7}{10} = 0{,}70$

Por tanto: $\frac{14}{25} < \frac{13}{20} < \frac{7}{10}$

2ª forma: Reducimos a común denominador:

$\frac{13}{20} = \frac{65}{100}$; $\frac{14}{25} = \frac{56}{100}$; $\frac{7}{10} = \frac{70}{100}$ Por tanto: $\frac{14}{25} < \frac{13}{20} < \frac{7}{10}$

8.- Calcula y simplifica:

a) $\frac{3}{5} \cdot \frac{2}{3}$ b) $6 \cdot \frac{3}{4}$ c) $5 : \frac{3}{4}$

d) $\frac{1}{3} \cdot \frac{4}{5}$ e) $\frac{8}{3} : \frac{2}{3}$ f) $\frac{2}{7} : 4$

a) $\frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}$ b) $6 \cdot \frac{3}{4} = \frac{18}{4} = \frac{9}{2}$ c) $5 : \frac{3}{4} = \frac{20}{3}$

d) $\frac{1}{3} \cdot \frac{4}{5} = \frac{4}{15}$ e) $\frac{8}{3} : \frac{2}{3} = \frac{24}{6} = 4$ f) $\frac{2}{7} : 4 = \frac{2}{28} = \frac{1}{14}$

9.- Calcula y simplifica:

a) $\frac{1}{2} + \frac{1}{4}$ b) $1 + \frac{1}{2}$ c) $2 - \frac{1}{4}$

d) $\frac{1}{2} - \frac{1}{4}$ e) $1 + \frac{1}{3}$ f) $\frac{1}{2} - \frac{1}{3}$

a) $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ b) $1 + \frac{1}{2} = \frac{3}{2}$ c) $2 - \frac{1}{4} = \frac{7}{4}$

d) $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ e) $1 + \frac{1}{3} = \frac{4}{3}$ f) $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$

10.- Calcula:

a) $\frac{1}{2} - \frac{1}{3} + \frac{1}{5}$ b) $\frac{5}{6} + \frac{1}{9} + \frac{3}{4}$

c) $\frac{1}{30} - \frac{1}{45}$ d) $\frac{11}{30} - \frac{3}{40} - \frac{7}{60}$

a) $\frac{1}{2} - \frac{1}{3} + \frac{1}{5} = \frac{15}{30} - \frac{10}{30} + \frac{6}{30} = \frac{11}{30}$

b) $\frac{5}{6} + \frac{1}{9} + \frac{3}{4} = \frac{30}{36} + \frac{4}{36} + \frac{27}{36} = \frac{61}{36}$

c) $\frac{1}{30} - \frac{1}{45} = \frac{3}{90} - \frac{2}{90} = \frac{1}{90}$

d) $\frac{11}{30} - \frac{3}{40} - \frac{7}{60} = \frac{44}{120} - \frac{9}{120} - \frac{14}{120} = \frac{21}{120} = \frac{7}{40}$

11.- Calcula:

a) $3 - \left(\frac{1}{6} + \frac{2}{3}\right)$ b) $\left(2 - \frac{2}{3}\right) + \left(5 - \frac{7}{2}\right)$

c) $\frac{3}{2} - 2 + \frac{1}{3}$ d) $5 - \left(\frac{1}{3} - 2\right)$

a) $3-\left(\frac{1}{6}+\frac{2}{3}\right)=\frac{3}{1}-\frac{1}{6}-\frac{2}{3}=\frac{18}{6}-\frac{1}{6}-\frac{4}{6}=\frac{13}{6}$

b) $\left(2-\frac{2}{3}\right)+\left(5-\frac{7}{2}\right)=\frac{4}{3}+\frac{3}{2}=\frac{8}{6}+\frac{9}{6}=\frac{17}{6}$

c) $\frac{3}{2}-2+\frac{1}{3}=\frac{9}{6}-\frac{12}{6}+\frac{2}{6}=\frac{-1}{6}$

d) $5-\left(\frac{1}{3}-2\right)=5-\left(\frac{-5}{3}\right)=5+\frac{5}{3}=\frac{20}{3}$

12.- Reduce a una sola fracción las expresiones:

a) $\frac{1}{2}-\frac{1}{4}\cdot\frac{1}{8}-\frac{1}{16}$

b) $\left(\frac{3}{5}-\frac{1}{4}+2\right)-\left(\frac{3}{4}-\frac{2}{5}+1\right)$

c) $\left(1+\frac{1}{3}\right)-\left(\frac{3}{4}+\frac{1}{2}\right)\cdot\left(\frac{1}{3}-\frac{1}{4}\right)$

d) $\left(\frac{3}{5}+\frac{1}{3}\right)-\left[1-\left(\frac{3}{4}-\frac{1}{2}\right)+\frac{2}{3}-\frac{3}{20}\right]$

a) $\frac{1}{2}-\frac{1}{4}\cdot\frac{1}{8}-\frac{1}{16}=\frac{1}{2}-\frac{1}{32}-\frac{1}{16}=\frac{16}{32}-\frac{1}{32}-\frac{2}{32}=\frac{13}{32}$

b) $\left(\frac{3}{5}-\frac{1}{4}+2\right)-\left(\frac{3}{4}-\frac{2}{5}+1\right)=\left(\frac{12}{20}-\frac{5}{20}+\frac{40}{20}\right)-\left(\frac{15}{20}-\frac{8}{20}+\frac{20}{20}\right)=$

$=\frac{47}{20}-\frac{27}{20}=\frac{20}{20}=1$

c) $\left(1+\frac{1}{3}\right)-\left(\frac{3}{4}+\frac{1}{2}\right)\cdot\left(\frac{1}{3}-\frac{1}{4}\right)=\frac{4}{3}-\frac{5}{4}\cdot\frac{1}{12}=\frac{4}{3}-\frac{5}{48}=\frac{64}{48}-\frac{5}{48}=\frac{59}{48}$

d) $\left(\frac{3}{5}+\frac{1}{3}\right)-\left[1-\left(\frac{3}{4}-\frac{1}{2}\right)+\frac{2}{3}-\frac{3}{20}\right]=$

$=\left(\frac{9}{15}+\frac{5}{15}\right)-\left[\frac{60}{60}-\frac{45}{60}+\frac{30}{60}+\frac{40}{60}-\frac{9}{60}\right]=$

$=\frac{14}{15}-\frac{76}{60}=\frac{14}{15}-\frac{19}{15}=\frac{-5}{15}=\frac{-1}{3}$

13.- Reduce:

a) $\frac{2}{3}\cdot\left(\frac{3}{4}-\frac{1}{2}\right)-\frac{1}{6}\cdot\left(\frac{5}{6}-\frac{1}{3}\right)$ b) $5:\left(\frac{2}{4}+1\right)-3:\left(\frac{1}{2}-\frac{1}{4}\right)$

a) $\frac{2}{3}\cdot\left(\frac{3}{4}-\frac{1}{2}\right)-\frac{1}{6}\cdot\left(\frac{5}{6}-\frac{1}{3}\right)=\frac{2}{3}\cdot\frac{1}{4}-\frac{1}{6}\cdot\frac{3}{6}=\frac{2}{3}\cdot\frac{1}{4}-\frac{1}{6}\cdot\frac{1}{2}=$

$=\frac{2}{12}-\frac{1}{12}=\frac{1}{12}$

b) $5:\left(\frac{2}{4}+1\right)-3:\left(\frac{1}{2}-\frac{1}{4}\right)=5:\frac{3}{2}-3:\frac{1}{4}=\frac{10}{3}-\frac{12}{1}=\frac{10}{3}-\frac{36}{3}=\frac{-26}{3}$

14.- Reduce a una fracción:

a) $\dfrac{1+\frac{1}{2}}{1-\frac{1}{2}}$ b) $\dfrac{3-\frac{5}{3}}{3+\frac{5}{3}}$ c) $\dfrac{\frac{1}{4}-\frac{3}{5}}{\frac{7}{10}-\frac{3}{4}}$

a) $\dfrac{1+\frac{1}{2}}{1-\frac{1}{2}}=\dfrac{\frac{3}{2}}{\frac{1}{2}}=3$

b) $\dfrac{3-\frac{5}{3}}{3+\frac{5}{3}}=\dfrac{\frac{4}{3}}{\frac{14}{3}}=\dfrac{4}{14}=\dfrac{2}{7}$

c) $\dfrac{\frac{1}{4}-\frac{3}{5}}{\frac{7}{10}-\frac{3}{4}}=\dfrac{\frac{5}{20}-\frac{12}{20}}{\frac{14}{20}-\frac{15}{20}}=\dfrac{\frac{-7}{20}}{\frac{-1}{20}}=7$